



Ingestible power capsule (Image courtesy: Massachusetts Institute of Technology)

can generate enough power to run small sensors or drug-delivery devices that can reside in the gastrointestinal tract for extended periods of time.

Acids, like those in the gastrointestinal tract, are known to carry electric currents. The researchers built a prototype device 40 millimetres (1.6 inches) in length and 12 millimetres (0.5 inch) in diameter. Attached to the surface of the device are electrodes made from zinc and copper foil. Inside the cylinder are a commercial temperature sensor and low-power wireless transmitter circuits.

This type of power could offer a safer and lower-cost alternative to the batteries currently used to power such devices, as per the researchers.

Solar-powered bicycle to combat air pollution

In an effort to reduce air pollution due to vehicular emissions in India, scientists have designed a solar-assisted bicycle that is pegged as a low-cost, eco-friendly alternative to two-wheeler motor vehicles. The bicycle can switch between the solar-energy-powered motor and pedals.

Major air pollutants emitted from vehicles are carbon monoxide (CO), nitrogen oxide (NOx) and particulate matter (PM). Increasing air pollution not only affects the environment, it also has adverse health effects like lung cancer and cardiovascular mortality, which may lead to death.

"To overcome this problem, an effort is being made for developing an eco-friendly vehicle to reduce pollution in India," says S.A. Puviyarasu from Dr N G P Institute of Technology, Anna University in Tamil Nadu. "If we use a solar-assisted bicycle, we can reduce 60 per cent of all vehicular pollution in the country," Puviyarasu adds.

The design consists of a brushless DC motor mounted on the front wheel of the bicycle, an electric throttle for varying the speed of the bicycle and a lead-acid battery that stores solar energy. A solar panel (of any specification)

is mounted on the bicycle carrier. Depending upon the specification of the solar panel, energy is produced.

When the power supply is given to the hub motor from the solar panel unit, the bicycle wheels rotate to move. The rider can choose between the motor and the pedals, or use both at the same time.

Device that turns air pollution into printing ink

Graviky Labs is an MIT Media Lab spin-off based in India, and it has come up with an ingenious solution to air pollution problems in Asia—convert polluted air into high-quality printing ink.



It takes 45 minutes worth of vehicular emissions captured by KAALINK to produce 29.6 millilitres (1 fluid ounce) of ink (Image courtesy: www.livescience.com)

While cheaper carbon black inks are manufactured through the deliberate burning of fossil fuels, Graviky Labs uses their proprietary device, KAALINK, to capture soot that is already being emitted from vehicles. KAALINK is retrofitted to the exhaust pipe of vehicles/generators to capture outgoing pollutants.

The collected soot undergoes various proprietary processes to remove heavy metals and carcinogens. The end product is a purified carbon-rich pigment. The carbon is then used to make different types of inks and paints.

World's first self-driving Tesla taxis to hit the roads in Dubai

The government of United Arab Emirates has purchased 200 Tesla vehicles to add to the fleet of Dubai Taxi Corp. fleet. Though the combination of Model S sedans and Model X SUVs will initially be used in autopilot mode, which requires a human driver, these will come equipped with the hardware needed for complete self-driving capability. The Roads and Transport Authority (RTA) in Dubai plans to do a test run in the autonomous driving mode, with the goal of making Dubai the smartest city by 2020.